

Reviewer Pack — Minimal Hodge Module Rank and DPLL Search Tree Size Correlat...

Entry c6d096039c04 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Module Rank and DPLL Search Tree Size Correlation

Entry ID: c6d096039c04 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 18:44:54 UTC

1. Conjecture

Field A (mathematical branch): Hodge Theory **Field B** (complexity object): DPLL Search Trees

Statement:

The minimal rank of the Hodge module associated with a Boolean circuit is linearly correlated with its DPLL search tree size, such that $\text{MinimalRank}(H) = \Theta(\text{SearchTreeSize})$.

Rationale (proposer's reasoning):

Hodge theory provides a framework for studying algebraic varieties, and its application to Boolean circuits could reveal hidden structures in their complexity. A constructive mapping from circuit structure to Hodge module rank might expose new insights into the nature of computational hardness.

Taxonomy category: HODGE_T0_DPLL (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 120f29e97cdd2bb5

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between $\text{MinimalRank}(H)$ and SearchTreeSize is ≥ 0.8 for all 30 circuits, and there are no outliers beyond a 95% confidence interval. The criterion is falsified if the Pearson correlation coefficient is < 0.5 or any seed produces a correlation coefficient > 1.2 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Hodge Module Rank AND DPLL Search Trees - Hodge Theory AND DPLL algorithm application - Boolean circuit AND Hodge module rank Θ (DPLL search tree size)

Top relevant hits considered: - [http://arxiv.org/abs/1204.1444v2] Computing the minimum rank of a loop directed tree - [http://arxiv.org/abs/1901.05780v2] Hodge filtration, minimal exponent, and local vanishing - [http://arxiv.org/abs/1801.10489v3] Hodge level for weighted complete intersections - [http://arxiv.org/abs/2006.03953v3] Hodge theory of degenerations, (II): vanishing cohomology and geometric applications - [http://arxiv.org/abs/2104.12645v2] Cohomological Descent for Faltings' p -adic Hodge Theory and Applications - [http://arxiv.org/abs/1109.2049v1] Structure-Based Local Search Heuristics for Circuit-Level Boolean Satisfiability - [http://arxiv.org/abs/2311.04204v3] Sharp Thresholds Imply Circuit Lower Bounds: from random 2-SAT to Planted Clique - [http://arxiv.org/abs/2406.16142v2] Nearly Optimal Circuit Size for Sparse Quantum State Preparation

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_circuit(n, m):
        variables = [f'x{i}' for i in range(n)]
        clauses = []
        for _ in range(m):
            clause = random.sample(variables, 2)
            if random.choice([True, False]):
                clause = [f'~{var}' if var.startswith('~') else f'{var}' for var in clause]
            clauses.append(' | '.join(clause))
        return ' & '.join(clauses)

    def dpll(circuit):
        literals = set()
        stack = []
        def solve():
            while True:
                if not stack:
                    if all(l in literals for l in circuit.split('&')):
                        return literals
                else:
```

```

        return None
    literal = next((l for l in literals if l not in circuit), None)
    if literal is None:
        literal = random.choice(list(literals))
    literals.add(literal)
    stack.append(literal)
    new_circuit = circuit.replace(f'{{literal}}', '').replace(f'-{{literal}}', '')
    result = solve()
    if result is not None:
        return result
    literals.remove(literal)
    stack.pop()
return solve()

def hodge_module_rank(circuit):
    # Placeholder for Hodge module rank computation
    # This is a dummy implementation and should be replaced with actual logic
    return len(circuit.split('&'))

n = random.randint(5, 30)
m = random.randint(n, 2 * n)
circuit = generate_circuit(n, m)
search_tree_size = len(dpll(circuit))
minimal_rank = hodge_module_rank(circuit)

return {
    "metric_name": "correlation",
    "metric_value": minimal_rank,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
        31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
        73, 79, 83, 89, 97, 101, 103, 107, 109, 113
    ]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
    elif any(r["metric_value"] > 1.2 for r in results):
        first_failing_seed = next((r["seed"] for r in results if r["metric_value"] > 1.2), None)
        print(f"RESULT: FALSIFIED counterexample='correlation_too_high' first_failing_seed={first_failing_seed}")
    else:

```

```
print(f"RESULT: INCONCLUSIVE reason=insufficient_evidence")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a8fa3b3e.py", line 83, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a8fa3b3e.py", line 62, in run_trial
    search_tree_size = len(dpll(circuit))
                       ^^^^^^^^^^^^^^^^^
```

TypeError: object of type 'NoneType' has no len()

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating the Pearson correlation coefficient and thus confirming or refuting the conjecture. | next: Review the test code for potential bugs that cause it to crash and rerun the test with a corrected version.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13906
2	preregistration	ollama_remote	glm4:latest	0	0	10378
3	novelty	ollama_remote	glm4:latest	0	0	8578
4	novelty	ollama_remote	glm4:latest	0	0	9707
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15567
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6701
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11251
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9338
9	judge	ollama_remote	glm4:latest	0	0	11895

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 97320 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/c6d096039c04.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c6d096039c04.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/c6d096039c04.tar.gz` (if generated)